Wenbo Ji

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Seeking Research Engineer / Research Scientist roles in video generation, multimodal generative models and world models.

Highlights

- **Video & controllable generation:** fine-tuned Wan2.1 in full (8×H100, 100k iterations) and with LoRA; conditioned video diffusion on reference images, 3DMM geometry, text and camera for identity-consistent portrait video.
- **Post-training & world models:** RL post-training (GRPO) for video generation; cross-embodiment video generation and video–action world models for robot dexterous manipulation.
- **Training & inference systems:** led deterministic CUDA/Triton kernels for MiniMax-H3 in RL-Kernel (attention **24–48×** faster on H100); Ulysses sequence parallelism in xDiT (6.59×) and Triton kernel fusion in FastVideo (1.155×).

Education

Technical University of Munich — Oct 2023 – Expected Mar 2027
M.Sc. Electrical Engineering and Information Technology — Grade: 1.7 (German scale; 1.0 = best)

Tongji University — Sep 2021 – Aug 2025
Master of Engineering, Computer Technology — GPA: 4.5/5.0

Nanjing Tech University — Sep 2017 – Jun 2021
B.Sc. Information and Computing Science — GPA: 3.5/4.0

Research Experience

TUM Visual Computing Group — Master's Thesis — Mar 2026 – Present

- **Project:** Camera-Controlled Human Motion Video Diffusion
- **Advisors:** Prof. Matthias Nießner, Yu Chi, Jiapeng Tang.
- **Research:** developing **camera-controlled video diffusion** for human motion and human–scene interaction under changing viewpoints, including **RL post-training**.

TUM Visual Computing Group — Guided Research — Apr 2025 – Mar 2026

- **Project:** ViDS: 3D Face Tracking-Driven Portrait Video Generation
- **Output & advisor:** first-author paper (arXiv preprint); advisor Prof. Matthias Nießner.
- **Method:** led ViDS, which uses Wan2.1 video diffusion as a **neural shader over 3DMM normal maps** for identity-consistent portrait animation with fine-grained expression control.
- **Results:** ranked first on **8 of 13 metrics** on VFHQ against four state-of-the-art methods, and on all four criteria of a 40-participant user study.

TUM Computer Vision Group — Guided Research — Feb 2024 – Aug 2025

- **Project:** CSG-Fusion: Feed-Forward Sparse-View 3D Gaussian Reconstruction
- **Output & advisors:** co-first-author paper (ICCV 2025 Workshop **Best Paper**); advisors Prof. Daniel Cremers, Yan Xia.
- **Method:** proposed and implemented CSG-Fusion, which removes the redundant, cross-view-inconsistent Gaussians of per-view feed-forward prediction via MASt3R **matching-based pruning**, **masked cross-attention** aggregation and **context-view supervision**; ran all experiments.
- **Results:** **+2.8 dB PSNR** over Splatt3R and LPIPS 0.231→0.186 on ScanNet++ (90% overlap); **about 24% fewer Gaussians** with 512×512 inputs; zero-shot evaluation on DTU.

Industry Experience

Agile Robots SE — Research Intern — Apr 2026 – Present

- **Project:** Video-Action World Models for Robot Dexterous Manipulation
- **Research:** leading **cross-embodiment video generation** that translates egocentric human demonstrations into robot-domain videos for policy learning, addressing data scarcity and the human–robot embodiment gap.

ImFusion GmbH — Research Intern — Aug 2024 – Apr 2025

- **Project:** LiteTracker: Low-Latency Online Tissue Tracking
- **Output & role:** second-author paper (MICCAI 2025); implemented causal online inference and EMA-flow initialization and ran the tracking evaluation.
- **Method:** replaced CoTracker3 Online's sliding window with frame-by-frame **causal inference** (a ring-buffer cache of correlation features) and **EMA-flow initialization**, cutting refinement from four passes to one without retraining.

- **Paper results: 29.67 ms P95 latency** for 1,024 points on one RTX 3090, about **1/7** of CoTracker3 Online and 1/2 of Track-On(12), with the best occlusion accuracy on SuPer (**97.45**).

Selected Publications

* Equal contribution † Corresponding author

ViDS: Video Diffusion Shader using 3D Face Tracking arXiv 2026
Wenbo Ji, Davide Davoli, Zhe Chen, Liam Schoneveld, Matthias Nießner, Jiapeng Tang†

CSG-Fusion: Consistent Sparse-View Gaussian Splatting via Matching-based Fusion ICCV E2E3D Workshop 2025
Yan Xia*†, **Wenbo Ji***, Weirong Chen, Daniel Cremers **Best Paper**

LiteTracker: Leveraging Temporal Causality for Accurate Low-latency Tissue Tracking MICCAI 2025
Mert Asim Karaoglu†, **Wenbo Ji**, Ahmed Abbas, Nassir Navab, Benjamin Busam, Alexander Ladikos

RE0: Recognize Everything with 3D Zero-shot Instance Segmentation ICRA 2025
Xiaohan Yan*, Zijian Jiang*, Yinghao Shuai*, Nan Wang, Xiaowei Song, **Wenbo Ji**, Ge Wu, Jinyu He, Gang Wei, Zhicheng Wang†

Selected Open-source Contributions

RL-Kernel Deterministic MiniMax-H3 operators · #483–#489, #493, #494, #576, #586 ★445

- **Method:** led a deterministic operator stack that keeps training and rollout **bitwise-identical** in MiniMax-H3 RL post-training: CUDA forward/backward kernels (timestep embedding, tri-modal AdaLN, fused RMSNorm, gated residuals) with TP/SP variants, and a deterministic **Triton flash-attention** backend.
- **Results:** bitwise-identical to Diffusers; outputs and gradients under 2/4/8-way sequence parallelism match a single GPU bitwise (8×B200); attention forward+backward **24–48×** faster (one H100).

xDiT Distributed inference for a 14B video DiT · #821 ★2.7k

- **Method:** added **Ulysses sequence parallelism** (on Wan's unified sequence-parallel attention, with uneven-token handling) and FSDP sharding to make 14B video DiT inference faster and lighter.
- **Results: 6.59×** faster on 8×B200 than stock Diffusers on one B200, with bit-identical output; FSDP cuts per-GPU memory from 46.6 GB to **14.2 GB**.

FastVideo Video generation training/inference · #1900, #1899 ★4.8k

- **Method:** #1900 fused the sparse-attention scatter write and four transposes into one **Triton kernel**; #1899 retuned the tiling of quantization-aware attention training (Attn-QAT).
- **Results: 1.155×** end-to-end Wan2.1-VSA-T2V-14B inference (one B200, bit-identical frames); **10.9%** lower Wan2.1-1.3B training-step latency (2×B200).

Technical Skills

Generative modeling: diffusion/DiT, flow matching, video generation (Wan2.1), multi-condition CFG.
Training & post-training: full fine-tuning, LoRA, RL post-training (GRPO).
ML systems: Python, PyTorch, Diffusers; FSDP and Ulysses sequence parallelism for training/inference; NVIDIA H100.
GPU programming: CUDA, Triton; kernel fusion, tiling, deterministic (bitwise-reproducible) kernels.
3D vision: 3D Gaussian Splatting, MASt3R/Splatt3R, 3DMM/FLAME, point tracking (CoTracker3).
Languages: Chinese (native), English (CET-6), German (B2).

Awards & Scholarships

Best Paper Award, ICCV 2025 E2E3D Workshop (CSG-Fusion, co-first author) 2025
DAAD Scholarship (German Academic Exchange Service) 2023 – 2024
National Encouragement Scholarship (China) 2019